

FINITE ELEMENT FORMULATION OF A TAPERED TIMOSHENKO BEAM FOR FREE LATERAL VIBRATION ANALYSIS

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A finite element model is developed for free lateral vibration analyses of linearly tapered Timoshenko beams. The shape functions are obtained from the homogeneous solution of the governing equations for static deflections. Illustrative examples demonstrate the performance of this element, and allow comparisons of results with those obtained by elements based on different formulations.

1. INTRODUCTION

Because of their completeness and ease of manipulation, polynomials are often used in finite element formulations. A deviation from this practice will pay dividends if improved accuracies can be obtained by using shape functions other than polynomials. This is the case when the homogeneous solution of the pertinent differential equation is available for development of the element matrices. It is not difficult to show that the use of the homogeneous solution of the differential equation, in a finite element formulation, will yield exact element matrices [1]. In turn, this implies that the computed nodal quantities, for the assembled set of elements, will also be exact.

For static analysis, use of the homogeneous solution of the differential equation yields the exact stiffness matrix and load vector for an element. For dynamic analysis, a similar procedure, leading to a dynamic stiffness matrix, may be used. However, in this case, the homogeneous solution, and hence also the dynamic stiffness matrix, become frequency dependent. Such frequency dependent matrices are awkward to deal with computationally. A compromise may be made by using the homogeneous solution of the static problem for development of the element mass matrix. In this case, the mass matrix will be approximate, while the stiffness matrix remains exact. The small loss in accuracy due to the approximate form of the mass matrix is amply compensated for by the resulting linear eigenvalue equation which is far easier to handle computationally. This approach is adopted in this paper.

Other investigators have presented finite element formulations for tapered beam elements [2–4]. In particular, To [4] has provided element matrices for a tapered Timoshenko beam. A cubic polynomial was employed for the deflection distribution, and a linear distribution for shear strain.

The present formulation differs in some important aspects with that presented by To. First, the stiffness matrix derived here is exact. Second, at each node To defined three quantities, namely, lateral displacement, cross-sectional rotation and shear strain. The energy expressions for the Timoshenko beam show that only the displacements and rotations need to be continuous from one element to the next: that is, the continuity of the shear strain is superfluous. Indeed, at the connection of two elements with two different cross-sectional areas, shear strains must not be made continuous, for that will violate the nodal equilibrium of the shear forces, and this point must be respected in using To's formulation. It is, of course, possible to leave the shear strains unconnected at all nodes. However, this will lead to an unnecessarily large system of equations. These difficulties may be overcome by using the lateral displacement and rotation as the nodal variables, as has been done in the present formulation.

2. ESSENTIAL EQUATIONS

The relevant equations for free vibrations of a tapered Timoshenko beam are as follows [5]: strain energy (a list of nomenclature is given in Appendix B),

$$V_e = \frac{1}{2} \int_0^L [EI(\psi')^2 + kAG(y' - \psi)^2] dx; \quad (1)$$

complementary kinetic energy,

$$T_e^* = \frac{\rho}{2} \int_0^L (A\dot{y}^2 + I\dot{\psi}^2) dx. \quad (2)$$

The governing differential equations of the beam may be obtained by using Hamilton's Principle: that is,

$$\delta \int_{t_1}^{t_2} (T_e^* - V_e) dt = 0. \quad (3)$$

This leads to

$$kAG(y' - \psi) + (EI\psi')' - \rho I\ddot{\psi} = 0, \quad [kAG(y' - \psi)]' - \rho A\ddot{y} = 0, \quad (4, 5)$$

and the boundary conditions

$$\text{either } \psi' = 0 \text{ or } \psi = 0, \quad \text{either } y' - \psi = 0 \text{ or } y = 0, \quad (6a, b)$$

To derive the desired shape functions, one solves equations (4) and (5) for the case, $\ddot{y} = \ddot{\psi} = 0$. Their solution may be expressed as

$$\begin{Bmatrix} y \\ \psi \end{Bmatrix} = [B]\{d\}, \quad (7)$$

where

$$[B] = \begin{bmatrix} \frac{A_0}{A_1} \left(\frac{E}{6b^2kG} + \frac{2}{A_1^2} \right) \ln z + \frac{x}{A_1} \left(\frac{x}{A} - \frac{2}{A_1} \right), & \frac{x^2}{A_0A}, & x, & 1 \\ -\frac{x^2}{A^2}, & \frac{x}{A} \left(\frac{1}{A} + \frac{1}{A_0} \right), & 1, & 0 \end{bmatrix}, \quad (8a)$$

$$\{d\} = [d_1, d_2, d_3, d_4]^T, \quad z = A/A_0. \quad (8b, c)$$

The coefficients in the vector $\{d\}$ may be related to the nodal variables (see Figure 1), $y_{x=0} = y_1$, $\psi_{x=0} = \psi_1$, $y_{x=L} = y_2$, $\psi_{x=L} = \psi_2$, resulting in

$$\{d\} = [F]\{\phi_e\}, \quad (9)$$

where

$$[F]^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ \frac{A_0}{A_1} \left(\frac{E}{6b^2kG} + \frac{2}{A_1^2} \right) \ln z_2 + \frac{L}{A_1} \left(\frac{L}{A_2} - \frac{2}{A_1} \right) & \frac{L^2}{A_0A_2} & L & 1 \\ -\frac{L^2}{A_2^2} & \frac{L}{A_2} \left(\frac{1}{A_2} + \frac{1}{A_0} \right) & 1 & 0 \end{bmatrix}, \quad (10a)$$

$$A_2 = A_0 + A_1L, \quad z_2 = A_2/A_0, \quad \{\phi_e\} = [y_1, \psi_1, y_2, \psi_2]^T. \quad (10b, c, d)$$

Equation (7) may be employed to develop an exact stiffness matrix, through the expression for the strain energy, as follows:

$$[K_e] = [F]^T[P][F], \quad [P] = \begin{bmatrix} P(1,1) & P(1,2) & 0 & 0 \\ & P(2,2) & 0 & 0 \\ & & 0 & 0 \\ \text{sym.} & & & 0 \end{bmatrix}, \quad (11a, b)$$

$$P(1,1) = \frac{EA_0^2}{3b^2} \left(-\frac{L^2}{2A_1A_2^2} + \frac{\ln z_2}{A_1^3} - \frac{L}{A_1^2A_2} \right) + \frac{kG}{A_1} \left(\frac{EA_0}{6b^2kG} \right)^2 \ln z_2, \quad (11c)$$

$$P(1,2) = -\frac{EA_0L^2}{6b^2A_2^2}, \quad P(2,2) = \frac{EA_0L}{6b^2A_2} \left(\frac{1}{A_2} + \frac{1}{A_0} \right). \quad (11d, e)$$

In the present investigation, the same shape functions are to be employed to develop an approximate mass matrix through the complementary kinetic energy, as

$$[M_e] = [F]^T[R][F], \quad (12)$$

where

$$[R] = \begin{bmatrix} R(1,1) & R(1,2) & R(1,3) & R(1,4) \\ & R(2,2) & R(2,3) & 0 \\ & & R(3,3) & 0 \\ \text{sym.} & & & R(4,4) \end{bmatrix}, \quad (13a)$$

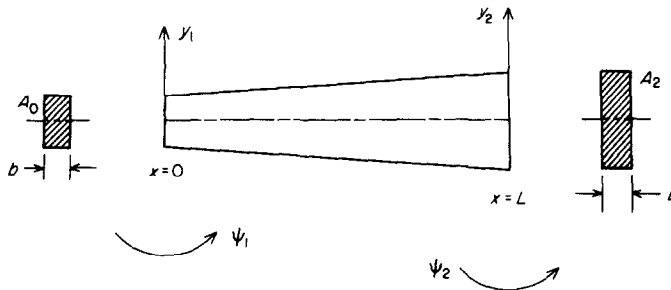


Figure 1. Tapered Timoshenko beam finite element.

$$\begin{aligned}
R(1, 1) = & \frac{1}{12b^2} \left(\frac{L^4}{4A_1} + \frac{A_0^4}{A_1^5} \ln z_2 - \frac{A_0 L^3}{3A_1^2} + \frac{A_0^2 L^2}{2A_1^3} - \frac{A_0^3 L}{A_1^4} \right) \\
& + \frac{A_0^2}{A_1^6} (A_1^4 \theta^2 + 4A_1^2 \theta + 4) \frac{A_2^2}{2A_1} (\ln z_2)^2 + \frac{A_0^2 \theta^2}{A_1^2} \left[-\frac{A_2^2}{2A_1} \ln z_2 + \frac{(A_0 + A_2)L}{4} \right] \\
& + \theta \left\{ -\frac{2A_0}{A_1^5} \left(\frac{A_0^3}{3} + 2A_0^2 A_1 L + 2A_0 A_1^2 L^2 + \frac{A_1^3 L^3}{3} \right) \ln z_2 + \frac{2A_0^3 L}{3A_1^4} + \frac{5A_0^2 L^2}{3A_1^3} - \frac{2A_0 L^3}{9A_1^2} \right\} \\
& - \frac{2A_0}{A_1^7} \left(-\frac{5A_0^3}{6} + 2A_0^2 A_1 L + 3A_0 A_1^2 L^2 + \frac{2}{3} A_1^3 L^3 \right) \ln z_2 \\
& - \frac{5A_0^3 L}{3A_1^6} + \frac{17A_0^2 L^2}{6A_1^5} + \frac{5A_0 L^3}{9A_1^4} + \frac{L^4}{4A_1^3}, \tag{13b}
\end{aligned}$$

$$\begin{aligned}
R(1, 2) = & \frac{1}{12b^2} \left(\frac{A_0^3}{A_1^4} \ln z_2 - \frac{L^3}{3A_1} + \frac{A_0 L^2}{2A_1^2} - \frac{A_0^2 L}{A_1^3} - \frac{L^4}{4A_0} \right) \\
& + \frac{\theta}{3A_1} \left(\frac{A_1^3 L^3 + A_0^3}{A_1^3} \ln z_2 - \frac{A_0^2 L}{A_1^2} + \frac{A_0 L^2}{2A_1^3} + \frac{L^3}{3} \right) \\
& + \frac{2A_1^3 L^3 + 5A_0^3}{3A_1^3} \ln z_2 - \frac{5A_0^2 L}{3A_1^5} + \frac{5A_0 L^2}{6A_1^4} - \frac{L^3}{9A_1^3} - \frac{L^4}{4A_0 A_1^2}, \tag{13c}
\end{aligned}$$

$$\begin{aligned}
R(1, 3) = & -\frac{L^3}{12b^2} \left(\frac{A_0}{3} + \frac{A_1 L}{4} \right) + A_0 \theta \left\{ \left(\frac{A_0 A_1^2 L^2}{2} + \frac{1}{3} A_1^3 L^3 - \frac{A_0^3}{6} \right) \frac{\ln z_2}{A_1^3} \right. \\
& + \frac{A_0^2 L}{6A_1^2} - \frac{A_0 L}{12A_1} + \frac{L^3}{9} \left. \right\} + \frac{A_0}{A_1^5} \left(-\frac{A_0^3}{3} + A_0 A_1^2 L^2 + \frac{2}{3} A_1^3 L^3 \right) \ln z_2 \\
& + \frac{A_0^3 L}{3A_1^4} - \frac{A_0^2 L^2}{6A_1^3} - \frac{4A_0 L^3}{9A_1^2} - \frac{L^4}{4A_1}, \tag{13d}
\end{aligned}$$

$$\begin{aligned}
R(1, 4) = & A_0 \theta \left\{ \left(\frac{A_0^2}{2} + A_0 A_1 L + \frac{A_1^2 L^2}{2} \right) \frac{\ln z_2}{A_1^2} - \frac{A_0 L}{2A_1} - \frac{L^2}{4} \right\} \\
& + \frac{A_0}{A_1^4} (A_0^2 + 2A_0 A_1 L + A_1^2 L^2) \ln z_2 - \frac{A_0^2 L}{A_1^3} - \frac{3A_0 L^2}{2A_1^2} - \frac{L^3}{3A_1}, \tag{13e}
\end{aligned}$$

$$\begin{aligned}
R(2, 2) = & \frac{1}{12b^2} \left(\frac{A_0^2}{A_1^3} \ln z_2 + \frac{L^2}{2A_1} - \frac{A_0 L}{A_1^2} + \frac{L^3}{A_0} + \frac{A_1 L^4}{4A_0^2} \right) \\
& + \frac{1}{A_0^2} \left(\frac{L^4}{4A_1} + \frac{A_0^4}{A_1^5} \ln z_2 - \frac{A_0 L^3}{3A_1} + \frac{A_0^2 L^2}{2A_1^3} - \frac{A_0^3 L}{A_1^4} \right), \tag{13f}
\end{aligned}$$

$$R(2, 3) = \frac{1}{12b^2} \left(A_0 L^2 + A_1 L^3 + \frac{A_1^2 L^4}{4A_0} \right) + \frac{L^4}{4A_0}, \quad R(2, 4) = \frac{L^3}{3A_0}, \tag{13g, h}$$

$$R(3, 3) = \frac{(A_2^2 + A_0^2)(A_2 + A_0)L}{48b^2} + L^3 \left(\frac{A_0}{3} + \frac{A_1 L}{4} \right), \tag{13i}$$

$$R(3, 4) = L^2 \left(\frac{A_0}{2} + \frac{A_1 L}{3} \right), \quad R(4, 4) = L \left(A_0 + \frac{A_1 L}{2} \right), \quad \theta = \frac{E}{6b^2 kG}. \tag{13j-l}$$

The form of these equations leads to a singularity for a uniform beam, when $A_1 = 0$. However, these singularities are removable, and there are finite limits as A_1 approaches zero. When dealing with uniform beams, these equations must be modified by replacing the logarithmic terms with Taylor series in powers of A_1 , as obtained in Appendix A.

3. ILLUSTRATIVE EXAMPLES

Two examples of employing the element presented in this paper are to be presented. The following parameters were employed in both examples: $E = 210$ GPa, $G = 80$ GPa, $b = 2.54$ cm, $k = 0.667$; cross-sectional area at left end of beam $= 6.45$ cm², beam length $= 25.4$ cm, cross-sectional area at right end of beam $= 0.50 A_0$.

3.1. EXAMPLE 1: FIXED-FIXED BEAM

The structure considered is illustrated in Figure 2(a). In Figures 2(b) and 2(c) are shown the first two natural frequencies as a function of the number of degrees of freedom, when incorporating different numbers of equal length finite elements. Also shown in Figures 2(b) and (c) are the results obtained by the element developed by To [4].[†] Further results of natural frequencies are listed in Table 1 for the first five modes. The results in these figures and table generally show a superior rate of convergence of the natural frequency for the present element. The first five mode shapes of the beam, when modelled with eight elements of the present formulation, are shown in Figure 3.

TABLE 1

Natural frequencies of fixed-fixed tapered Timoshenko beam with area ratio 0.5;
(a) Cleghorn/Tabarrok; (b) To

(a)

No. of elements	No. of global degrees of freedom	1	2	3	4	5
2	2	0.93856E+04	0.34163E+05	—	—	—
3	4	0.92332E+04	0.25050E+05	0.59150E+05	0.11991E+06	—
4	6	0.91972E+04	0.24561E+05	0.47191E+05	0.91471E+05	0.15040E+06
5	8	0.91845E+04	0.24359E+05	0.46219E+05	0.74565E+05	0.12980E+06
6	10	0.91788E+04	0.24261E+05	0.45701E+05	0.72990E+05	0.10616E+06
7	12	0.91758E+04	0.24208E+05	0.45389E+05	0.72051E+05	0.10389E+06
8	14	0.91740E+04	0.24176E+05	0.45195E+05	0.71395E+05	0.10248E+06
9	16	0.91728E+04	0.24155E+05	0.45067E+05	0.70944E+05	0.10138E+06
10	18	0.91720E+04	0.24141E+05	0.44979E+05	0.70628E+05	0.10057E+06
11	20	0.91714E+04	0.24131E+05	0.44916E+05	0.70398E+05	0.99963E+05
12	22	0.91710E+04	0.24123E+05	0.44869E+05	0.70226E+05	0.99506E+05
13	24	0.91707E+04	0.24118E+05	0.44834E+05	0.70094E+05	0.99152E+05

(b)

No. of elements	No. of global degrees of freedom	1	2	3	4	5
2	5	0.98041E+04	0.31826E+05	0.14705E+06	0.17685E+06	0.50002E+06
3	8	0.93004E+04	0.25485E+05	0.52918E+05	0.10772E+06	0.23589E+06
4	11	0.92124E+04	0.24491E+05	0.47378E+05	0.78955E+05	0.13413E+06
5	14	0.91872E+04	0.24252E+05	0.45587E+05	0.74191E+05	0.10834E+06
6	17	0.91779E+04	0.24167E+05	0.45077E+05	0.71284E+05	0.10475E+06
7	20	0.91738E+04	0.24130E+05	0.44873E+05	0.70323E+05	0.10061E+06
8	23	0.91718E+04	0.24112E+05	0.44775E+05	0.69910E+05	0.98992E+05

[†] In reference [4], the expression for K_{55} has a typographical error. The multiplying factor should be HL^2 .

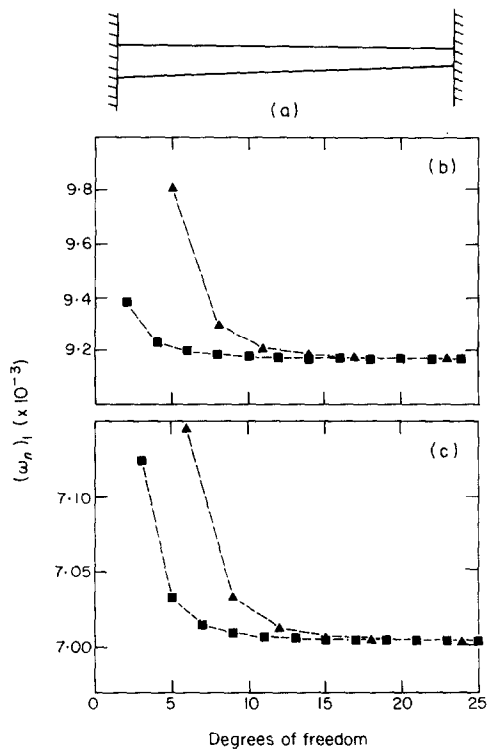


Figure 2. Fixed-fixed tapered Timoshenko beam with area ratio 0.5. (a) Beam; (b) first natural frequency as a function of the number of degrees of freedom; (c) second natural frequency as a function of the number of degrees of freedom. ■, Cleghorn/Tabarrok; ▲, To.

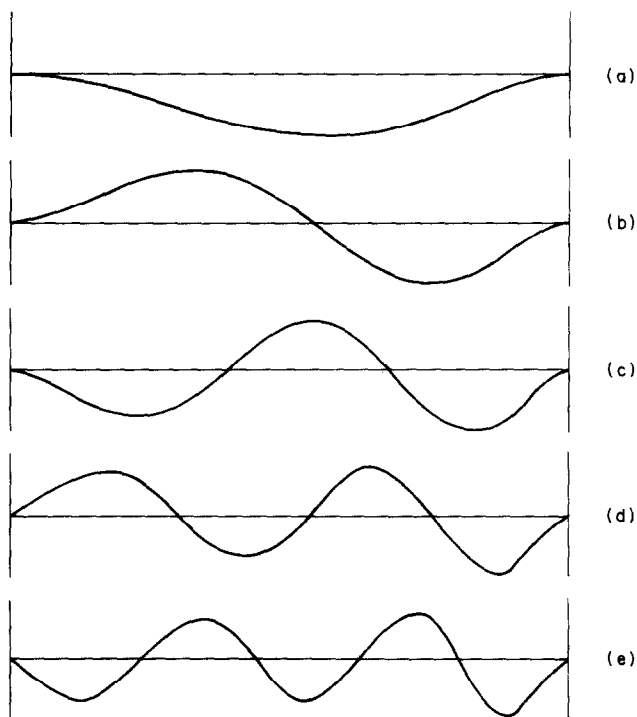


Figure 3. Mode shapes of fixed-fixed tapered Timoshenko beam with area ratio 0.5. (a) First mode; (b) second mode; (c) third mode; (d) fourth mode; (e) fifth mode.

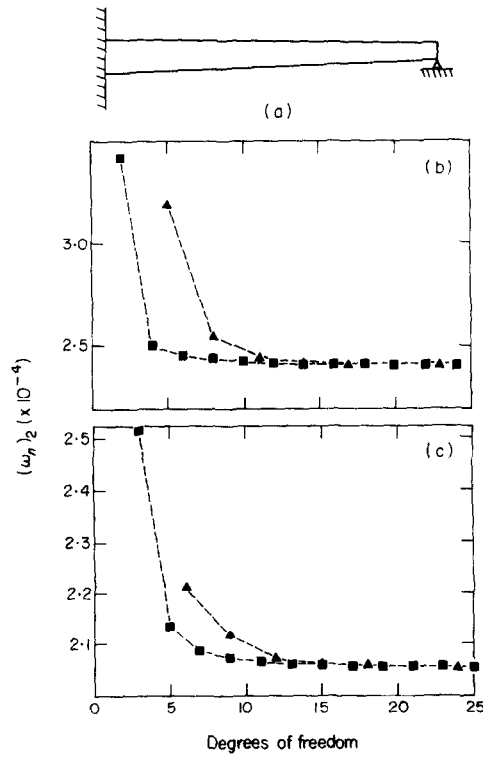


Figure 4. Fixed-pinned tapered Timoshenko beam with area ratio 0.5. (a) Beam; (b) first natural frequency as a function of the number of degrees of freedom; (c) second natural frequency as a function of the number of degrees of freedom. Key as Figure 2.

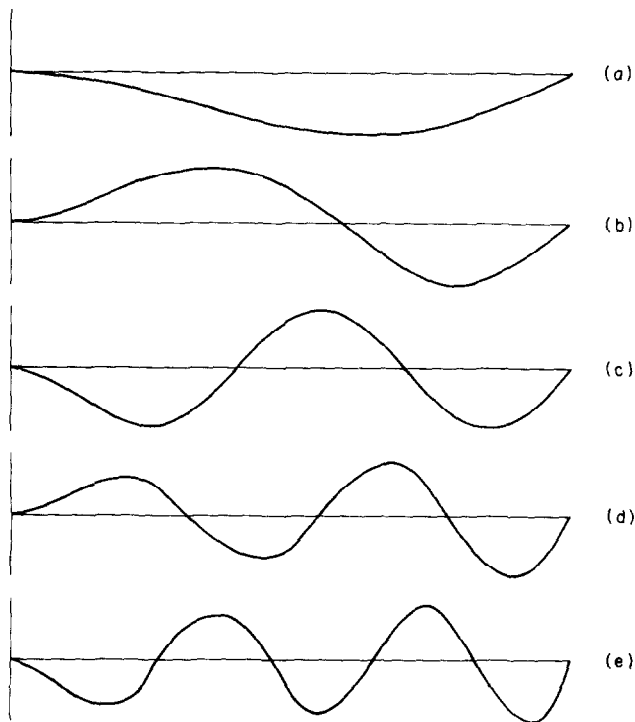


Figure 5. Mode shapes of fixed-pinned tapered Timoshenko beam with area ratio 0.5. (a) First mode; (b) second mode; (c) third mode; (d) fourth mode; (e) fifth mode.

TABLE 2

Natural frequencies of fixed-pinned tapered Timoshenko beam with area ratio 0.5;
(a) *Cleghorn/Tabarrok*; (b) *To*

(a)

No. of elements	No. of global degrees of freedom	1	2	3	4	5
1	2	0.22478E+04	0.19299E+05	—	—	—
2	4	0.22217E+04	0.10459E+05	0.32698E+05	0.87330E+05	—
3	6	0.22185E+04	0.10410E+05	0.26158E+05	0.57809E+05	0.10815E+06
4	8	0.22178E+04	0.10371E+05	0.26016E+05	0.48163E+05	0.90069E+05
5	10	0.22176E+04	0.10355E+05	0.25860E+05	0.47813E+05	0.75466E+05
6	12	0.22175E+04	0.10347E+05	0.25765E+05	0.47449E+05	0.74762E+05
7	14	0.22175E+04	0.10343E+05	0.25709E+05	0.47175E+05	0.74131E+05
8	16	0.22174E+04	0.10340E+05	0.25674E+05	0.46990E+05	0.73574E+05
9	18	0.22174E+04	0.10339E+05	0.25651E+05	0.46862E+05	0.73157E+05
10	20	0.22174E+04	0.10337E+05	0.25635E+05	0.46772E+05	0.72850E+05
11	22	0.22174E+04	0.10337E+05	0.25623E+05	0.46705E+05	0.72622E+05
12	24	0.22174E+04	0.10336E+05	0.25614E+05	0.46656E+05	0.72448E+05

(b)

No. of elements	No. of global degrees of freedom	1	2	3	4	5
2	6	0.22181E+04	0.10511E+05	0.29329E+05	0.69242E+05	0.17195E+06
3	9	0.22176E+04	0.10365E+05	0.26489E+05	0.52432E+05	0.96725E+05
4	12	0.22175E+04	0.10345E+05	0.25826E+05	0.49010E+05	0.80223E+05
5	15	0.22174E+04	0.10338E+05	0.25677E+05	0.47313E+05	0.77010E+05
6	18	0.22174E+04	0.10336E+05	0.25623E+05	0.46822E+05	0.73746E+05
7	21	0.22174E+04	0.10335E+05	0.25600E+05	0.46625E+05	0.72639E+05
8	24	0.22174E+04	0.10334E+05	0.25588E+05	0.46532E+05	0.72155E+05

3.2. EXAMPLE 2: FIXED-PINNED BEAM

The structure considered is illustrated in Figure 4(a). In Figures 4(b) and 4(c) are shown the first two natural frequencies when incorporating different numbers of equal length finite elements. Further results are listed in Table 2 for the first five modes. The first five mode shapes of the beam, when modelled with four elements of the present formulation, are shown in Figure 5.

4. CONCLUSIONS

A finite element has been formulated for tapered Timoshenko beams. The example problems presented show that this element out-performs an earlier developed element.

ACKNOWLEDGMENTS

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APPENDIX A: ALTERNATE FORM OF ELEMENT MATRIX COMPONENTS

$$\begin{aligned}
 R(1, 1) = & \frac{A_0^2 \theta^2}{2} \{ A_0 L Q_2 - A_0^2 Q_2 R_1 + (A_0 + A_2) L R_1^2 - A_0^2 Q_3 - L^2 R_1 \} \\
 & + \theta \left\{ 2A_0^3 L Q_4 - A_0^2 L^2 Q_3 - 2A_0^4 R_1 Q_4 + 4A_0^3 L R_1 Q_3 - 2A_0^2 L^2 R_1 Q_2 - \frac{2A_0^4 Q_5}{3} \right\} \\
 & - \frac{L^4 Q_3}{6} - \frac{L^5 Q_2}{15A_0} + 2A_0^3 L Q_6 - 3A_0^2 L^2 Q_5 + \frac{2A_0 L^3 Q_4}{3} - 2A_0^4 R_1 Q_6 + 4A_0^3 L R_1 Q_5 \\
 & - 2A_0^2 L^2 R_1 Q_4 + \frac{5A_0^4 Q_7}{3} + \frac{A_0^4 Q_5}{12b^2}, \quad (A1)
 \end{aligned}$$

$$R(1, 2) = \frac{\theta}{3} (L^3 R_1 - A_0^3 Q_4) + \frac{2L^3 Q_3}{3} - \frac{5A_0^3 Q_6}{3} - \frac{1}{12b^2} \left(A_0^3 Q_4 + \frac{L^4}{4A_0} \right), \quad (A2)$$

$$\begin{aligned}
 R(1, 3) = & \frac{A_0^2 \theta}{2} (L^2 R_1 - A_0^2 Q_3) + A_0^2 L^2 Q_3 - \frac{A_0^4 Q_5}{3} + \frac{A_0 \theta}{3} (L^3 A_1 R_1 - A_0^3 A_1 Q_4) \\
 & - \frac{2A_0 L^3 Q_2}{3} - \frac{L^3}{12b^2} \left(\frac{A_0}{3} + \frac{A_1 L}{4} \right), \quad (A3)
 \end{aligned}$$

$$\begin{aligned}
 R(1, 4) = & A_0^2 \theta \left(\frac{L^2}{A_0} - A_0 Q_2 + L R_2 \right) - A_0^3 Q_4 + 2A_0^2 L Q_3 \\
 & + \frac{A_0 \theta}{2} (L^2 A_1 R_1 - A_0^2 A_1 Q_3) - A_0 L^2 Q_2, \quad (A4)
 \end{aligned}$$

$$R(2, 2) = \frac{1}{12b^2} \left[A_0^2 Q_3 + \frac{2L^3}{3A_0} + \frac{L^3}{A_0^2} \left(\frac{A_0}{3} + \frac{A_1 L}{4} \right) \right] + A_0^2 Q_5, \quad (A5)$$

$$R(2, 3) = \frac{1}{6b^2} \left(A_0 L^2 + A_1 L^3 + \frac{A_1^2 L^4}{4A_0} \right) + \frac{L^4}{4A_0}, \quad (A6)$$

$$R(2, 4) = \frac{L^3}{3A_0}, \quad R(3, 3) = \frac{(A_2^2 + A_0^2)(A_2 + A_0)L}{48b^2} + L^3 \left(\frac{A_0}{3} + \frac{A_1 L}{4} \right), \quad (A7, A8)$$

$$R(3, 4) = L^2 \left(\frac{A_0}{2} + \frac{A_1 L}{3} \right), \quad (A9)$$

$$R(4, 4) = L \left(A_0 + \frac{A_1 L}{2} \right), \quad (\text{A10})$$

$$P(1, 1) = -\frac{A_1 L^4}{6b^2 A_2^2} + \frac{A_0^2}{3b^2} Q_2 + kG \left(\frac{EA_0}{6b^2 kG} \right) R_1, \quad (\text{A11})$$

$$[F]^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ \frac{EA_0}{6b^2 kG} R_1 - \frac{L^3}{A_0 A_2} + 2A_0 Q_3 & \frac{L^2}{A_0 A_2} & L & 1 \\ -\frac{L^2}{A_2^2} & \frac{L}{A_2} \left(\frac{1}{A_2} + \frac{1}{A_0} \right) & 1 & 0 \end{bmatrix}, \quad (\text{A12})$$

$$R_1 = \frac{L}{A_0} - \frac{A_1 L^2}{2A_0^2} + \dots = \sum_{i=1}^{\infty} \frac{(-A_1)^{i-1}}{i} \left(\frac{L}{A_0} \right)^i, \quad (\text{A13})$$

$$R_2 = -\frac{A_1 L^2}{2A_0^2} + \frac{A_1^2 L^3}{3A_0^3} - \dots \left(= R_1 - \frac{L}{A_0} \right), \quad (\text{A14})$$

$$R_3 = \frac{A_1^2 L^3}{3A_0^3} - \frac{A_1^3 L^4}{4A_0^4} + \dots, \quad (\text{A15})$$

$$-A_1 Q_2 = -A_1 \left(\frac{L^2}{2A_0^2} - \frac{A_1 L^3}{3A_0^3} + \frac{A_1^2 L^4}{4A_0^4} - \dots \right) = R_2, \quad (\text{A16})$$

$$A_1^2 Q_3 = A_1^2 \left(\frac{L^3}{3A_0^3} - \frac{A_1 L^4}{4A_0^4} + \frac{A_1^2 L^5}{5A_0^5} - \dots \right) = R_3, \quad (\text{A17})$$

$$-A_1^3 Q_4 = -A_1^3 \left(\frac{L^4}{4A_0^4} - \frac{A_1 L^5}{5A_0^5} + \frac{A_1^2 L^6}{6A_0^6} - \dots \right) = R_4. \quad (\text{A18})$$

APPENDIX B: NOMENCLATURE

A	$= A_0 + A_1 x$, cross-sectional area	k	dimensionless shape factor
b	width of cross-section	L	length of element
E	modulus of elasticity	y	centerline deflection of the beam
G	shear modulus	ρ	mass density
h	height of cross-section	ψ	rotation of cross-section
I	second moment of area of cross-section		